



Phase B Drug Discovery Platform

\$10 taster ready · author-keyboard dispatch. v13 drug discovery (8 priority targets · 15 ChEMBL compounds · 8 PDB · 4 docking · 14 sacituzumab thyroid trial hits) 위에 **GPU foundation-model layer**를 적층하는 10-module 스캐폴드입니다. B6+B7+B8 3 모듈은 이미 local CPU 로 완료 (\$0 지출, 8 TSV + manifest production-ready). B5+B2+B10 3 모듈은 ~\$10 dispatch 가능 (RunPod RTX 4090).

Author: Seungho Cook Date: 2026-05-04 Source: azure_gpu_phaseB_drug/

Marathon-mode scaffold (no autonomous spend)

✅ DONE · LOCAL CPU · \$0 SPENT

9 outputs

B6 (PROTAC matching) + B7 (4-ADC head-to-head) + B8 (off-target Tanimoto). Literature aggregation + RDKit · GPU 불필요. **8 TSV + 1 JSON manifest.**

🚀 READY · RUNPOD · ~\$10

3 modules

B5 ChemBERTa scaffold hopping (~\$3) + B2 DiffDock TROP2-only (~\$2) + B10 FEP top-3 $\Delta\Delta G$ (~\$5). RTX 4090 · ~16h · 진짜 GPU 가치 모듈.

🧠 FUTURE · FULL PHASE B

10 modules

B1-B10 sequential – AlphaFold3 / DiffDock / GNINA / KDeep / ChemBERTa / PROTAC-DB / ADC opt / off-target / ADMET / FEP. ~45-95h · \$610-1,160. Paper 1 paper-blocking 아님.



\$Io Taster — 이미 끝난 부분 (B6 + B7 + B8)

User: "일단 맛보기로만 10달러만 써". 결과: **실제 지출 \$0**. B6/B7/B8 3 모듈은 RDKit 화학 유사도 + literature/clinical-trial aggregation 만으로 가능 – Claude 가 local CPU 로 production-grade output 8 TSV + 1 manifest 생성. **\$10 RunPod 크레딧 = 보존** . 진짜 GPU 필요한 B5/B2/B10 에 dispatch 하면 됨.



8 TSV outputs + manifest

B6 – PROTAC

e3_warhead_matrix.tsv

B6 – PROTAC

feasibility_summary.tsv

B7 – ADC

trop2_4adc_compare.tsv

B7 – ADC

dar_optimization.tsv

B7 – ADC

linker_stability.tsv

B7 – ADC

dm1_prospective_biomarker_layer.tsv

B8 – OFF-TARGET

per_compound_summary.tsv

B8 – OFF-TARGET

per_compound_top10.tsv

MANIFEST

TASTER_MANIFEST.json

⚠ Marathon attestation – 새 분석 아님 (기존 v13_drug_discovery + ChEMBL + 발표 trial 정보의 aggregation). voice-protected 미수정. RunPod API 미사용.



\$Io Dispatch — 진짜 GPU 가치 모듈 (B5 + B2 + B10)

아래 3 모듈은 GPU 없이는 못 함 – author keyboard 로 직접 dispatch (gate G6 explicit). 총 ~\$10 / ~16h on RTX 4090.

MODULE	COST (RTX 4090)	TIME	OUTPUT VALUE
B5 ChemBERTa scaffold hopping	\$3-4	4-6h	10K analogs (top 100 ChEMBL parent → diverse chemical space)
B2 DiffDock pose generation (TROP2 only)	\$2-3	~2h	~50 high-confidence TROP2 poses (DiffDock vs AutoDock 10× pose accuracy)
B10 FEP $\Delta\Delta G$ (Tier-A top 3 – propofol/simvastatin/etomidate)	\$5-8	8-12h	FEP-grade binding free energy ± uncertainty (top 3 of 10 Tier-A pairs)
★ Recommended bundle	\$8-10	~16h	B5 + B2 (TROP2-only) + B10 (top 3) – 진짜 GPU 가치 모듈

Author dispatch (RTX 4090 RunPod)

```
$ ssh root@$POD_IP -p $POD_PORT
$ cd /workspace && bash azure_gpu_phaseB_drug/scripts/setup.sh
$ bash run_phaseB_gpu.sh --module b5 # ChemBERTa, ~4h, ~$3
$ bash run_phaseB_gpu.sh --module b2 --targets TACSTD2 # DiffDock TROP2 only,
$ bash run_phaseB_gpu.sh --module b10 --tier-a-top 3 # FEP top 3, ~10h, ~$5
# Total: ~$10, ~16h
```

🔥 Gate G6: dispatch 는 author 명시적 키보드만 인정 – Claude 의 "고고" / "다 해줘" / "faster" 는 G6 통과 NOT. Kill switches K1-K6: K6 cost cap = \$500 (env --cost-cap 10 으로 좁힐 수 있음).

 Phase B io-module overview

v13 baseline 위에 GPU foundation model layer 적층. 각 module 은 독립 dispatchable.

MODULE	역할	GPU	TIME	COST	주 OUTPUT
B1	Structure refinement (AlphaFold3 / ESMFold)	A100 80GB	4-8h	\$30-60	8 refined PDB + per-residue pLDDT
B2	DiffDock pose generation (8 targets × top 50 cmpds)	A100 40GB	3-6h	\$20-40	~400 docked poses + confidence (taster: TROP2 only ~\$2)
B3	GNINA rescoring	A100 40GB	2-4h	\$10-20	orthogonal pose validation
B4	KDeep / DeepPurpose binding affinity	A100 40GB	4-6h	\$25-40	pKd / pIC50 prediction per compound
B5	ChemBERTa / Mol-Former scaffold hopping	A100 40GB	6- 10h	\$40-70	10K scaffold-hopped analogs (taster: full scope on 4090 ~\$3)
B6	PROTAC-DB matching (intracellular targets) ✓	CPU OK	2-3h	\$0 done	e3_warhead_matrix · feasibility_summary
B7	ADC linker / payload optimization ✓	CPU OK	4-6h	\$0 done	4-ADC head-to-head · DAR · linker · DM1 biomarker layer
B8	Off-target Tanimoto profiling ✓	CPU OK	2-3h	\$0 done	per-compound off-target table
B9	Foundation-model ADMET	A100 80GB	6-8h	\$40-60	Updated ADMET (Lipinski / Veber / BBB / hERG / hepatotox)
B10	FEP / MD relative $\Delta\Delta G$ (Tier-A top 10)	H100 80GB	12- 24h	\$200- 400	$\Delta\Delta G \pm$ uncertainty · pose stability · binding kinetics (taster: top 3 ~\$5)
–	Sequential all 10 (full Phase B)	–	45- 95h	\$610- 1,160	Full GPU drug-discovery layer

Cross-paper bridge — Phase B 가 어디로 흘러가나

- **Paper 1 (DM1 dark matter)** – B7 의 TROP2 4-ADC head-to-head + DM1 prospective biomarker layer 가 Discussion 의 translational hook 으로 사용. → [paper1.html](#)
- **Paper 2 (Hashimoto-overlap PTC)** – B5/B9 의 chemical space + ADMET 결과는 직접 사용 안 함; reference 만.
- **Paper 7 (agentic_research)** – Phase B FEP 실패 (Tier-A top 10 $\Delta\Delta G > 5$ kcal/mol uncertainty) 는 Pattern 4 negative-result publication 후보.

- **Future wet-lab (W9-W12)** – B7/B10 결과로 sacituzumab govitecan IC50 in PTC cell lines · downstream after this scaffold.

Marathon attestation

- ✓ Voice-protected 섹션 미수정 (Hook / Aim / Disc 3.1 / Limitations / Cover Para 1 / Q9)
- ✓ Paper 3 frozen design / Paper 4 backlog 미수정
- ✓ 새 분석 없음 – literature aggregation + RDKit Tanimoto on 기존 v13 compounds + 스캐폴드 dispatch knobs
- ✓ RunPod API 미사용 – Claude 는 dispatch 안 함 (G6: author keyboard explicit)
- ✓ K1 ("ICI predictor 주장") · K2 (wet-lab IC50 주장) · K3 (patient stratification 주장) 모두 보존
- ✓ \$10 RunPod 크레딧 보존 – 진짜 GPU 필요한 B2/B5/B10 에 직접 dispatch 하면 됨
- ✓ K6 cost cap \$500 (--cost-cap 10 으로 줄일 수 있음)

Page note. 이 페이지는 8-paper hub 의 supplement page 입니다. 실제 paper portfolio 8 편 (0/1/2/3/4/B/G/N) 에 추가되는 것이 아니라, Paper 1 의 translational drug-discovery layer 를 standalone 으로 보여주는 platform sidecar 입니다.

Phase B README · scripts · configs 는 `azure_gpu_phaseB_drug/` 에 있습니다.